

The hand that holds still

A jar-opening puzzle for people building systems with more than one agent.

CONCEPTUAL TASK



Let the jar turn too, and motion may achieve nothing.

The stabilizing hand may move little while enabling the other hand to work.

Useful work includes keeping the wrong thing from moving.

Two jobs. One opening.

The arms act on different parts of the same object. These arrows show responsibilities, not measured forces.

STABILIZING ARM

Hold the jar body steady.

ACTING ARM

Grip and turn the lid.

JAR BODY

Resist unwanted rotation.

LID

Rotate relative to the jar.

VoxAct-B uses explicit roles, alongside language and voxels. Its results do not isolate roles alone.

The task depends on relative motion between lid and jar.

A fair rival can coordinate too.

A joint policy is not two independent learners competing over the same jar.

JOINT POLICY

Learn both arms together

ACT predicts action chunks from observations.

A policy can learn the relationship between both arms.

EXPLICIT ROLES

Name who holds and turns

VoxAct-B distinguishes acting and stabilizing arms.

Its full method also changes representation and uses language.

These published methods provide context. Cross-paper results are not the proposed matched comparison.

Do not weaken the baseline to make explicit roles look useful.

Invite the awkward third guest.

Proposed test: use one shared backbone and action interface. Then compare three conditions.

01

A / Joint

Control both arms from the same scene observations.

02

B / Roles

Add meaningful acting and stabilizing role information.

03

C / Constant

Keep B's module. Replace role input with one constant token in training and evaluation.

B vs C isolates role input within B; A still differs architecturally.

If B matches C, role information has not earned the credit.

Keep the budget. Keep the failures.

Extra information or computation could explain an apparent coordination advantage.

01

Match resources

Same demonstrations, observations and scenes. Common total compute caps, including role assignment.

02

Count rescues

A human rescue makes an assisted outcome. Retain timeouts and attempts that never start.

03

Show tradeoffs

Faster successes do not erase failures. Missing telemetry does not mean zero collisions.

Proposed comparison only. No robot, simulation or training run performed.

Report completion, collisions and interventions side by side.

Who keeps the jar from turning?

The software analogy is a question to test, not a result established by robot experiments.

A QUESTION FOR A BUILDER

What should the next agent do?

What must stay stable for the action to succeed?

For example: preserve a checked input while another component edits a draft. Compare the arrangement before claiming it helps.

Design analogy / Outcome unmeasured

Name the stabilizing responsibility before adding another busy participant.

Read the method. Leave the winner open.

Primary papers supply the context. The Daily Special supplies the unexecuted comparison.

S1 **VoxAct-B**

2024 / checked 28 Sep 2026

S2 **ACT / ALOHA**

2023 / checked 28 Sep 2026

S3 **Two arms, one jar**

2026-09-28 / checked 28 Sep 2026

The jar drawing is conceptual. No success percentages, torque values or measured software gains are implied.

Published robot demonstrations do not settle this role ablation.